

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/Downloads/hw5-natalie-evelyn5-Fall125`
Installed OpenSSL          v1.6.0
Installed GR_jll           v0.73.18+0
Installed OpenBLAS32_jll   v0.3.29+0
```

```

Installed PlotUtils          v1.4.4
Installed Measures           v0.3.3
Installed StaticArrays       v1.9.15
Installed MutableArithmetics v1.6.7
Installed HiGHS_jll          v1.12.0+0
Installed Pango_jll          v1.57.0+0
Installed JSON               v1.3.0
Installed FFMPEG              v0.4.5
Installed StaticArraysCore   v1.4.4
Installed DataStructures     v0.19.3
Installed Graphs             v1.13.1
Installed METIS_jll          v5.1.3+0
Installed GR                  v0.73.18
Installed StatsBase          v0.34.8
Installed GraphRecipes       v0.5.15
Installed ChainRulesCore     v1.26.0
Installed HiGHS              v1.20.1
Installed JuMP               v1.29.3
Installed Interpolations     v0.16.2
Installed SimpleTraits       v0.9.5
Installed Adapt             v4.4.0
Installed FFMPEG_jll         v8.0.0+0
Installed StableRNGs         v1.0.4
Installed StructUtils        v2.6.0
Installed ForwardDiff        v1.3.0
Precompiling project...
 558.8 ms Measures
 590.4 ms StaticArraysCore
 601.4 ms StableRNGs
 618.1 ms Adapt
 679.3 ms StructUtils
 658.0 ms SimpleTraits
 640.4 ms METIS_jll
1125.6 ms ChainRulesCore
 540.7 ms EarCut_jll
1925.6 ms DataStructures
 716.7 ms OpenBLAS32_jll
1046.4 ms ColorVectorSpace → SpecialFunctionsExt
2014.2 ms OpenSSL
1458.3 ms Pango_jll
 571.3 ms DiffResults
1525.0 ms FFMPEG_jll
 478.6 ms OffsetArrays → OffsetArraysAdaptExt

```

745.7 ms	Adapt → AdaptSparseArraysExt
655.6 ms	StructUtils → StructUtilsTablesExt
5093.7 ms	MutableArithmetics
4256.4 ms	ColorSchemes
735.7 ms	ChainRulesCore → ChainRulesCoreSparseArraysExt
1105.0 ms	LogExpFunctions → LogExpFunctionsChainRulesCoreExt
663.3 ms	SortingAlgorithms
1404.8 ms	SpecialFunctions → SpecialFunctionsChainRulesCoreExt
876.1 ms	HiGHS_jll
3316.1 ms	JSON
1972.9 ms	FFMPEG
6329.1 ms	StaticArrays
2358.6 ms	GR_jll
3499.2 ms	ForwardDiff
1369.5 ms	BenchmarkTools
2234.4 ms	StatsBase
770.9 ms	StaticArrays → StaticArraysChainRulesCoreExt
1309.0 ms	ArnoldiMethod
617.8 ms	StaticArrays → StaticArraysStatisticsExt
608.5 ms	Adapt → AdaptStaticArraysExt
973.2 ms	ForwardDiff → ForwardDiffStaticArraysExt
9038.5 ms	PlotUtils
1948.4 ms	GeometryTypes
16413.9 ms	HTTP
10872.3 ms	GeometryBasics
4309.8 ms	Graphs
2602.9 ms	Interpolations
2714.4 ms	PlotThemes
3615.5 ms	RecipesPipeline
1657.3 ms	NetworkLayout
3728.7 ms	GR
1248.5 ms	Interpolations → InterpolationsForwardDiffExt
1655.4 ms	NetworkLayout → NetworkLayoutGraphsExt
4457.1 ms	GraphRecipes
38844.8 ms	MathOptInterface
43660.3 ms	DataFrames
2630.0 ms	MathOptIIS
2696.2 ms	Latexify → DataFramesExt
8146.1 ms	HiGHS
15111.8 ms	JuMP
48983.8 ms	Plots
2943.0 ms	Plots → GeometryBasicsExt

59 dependencies successfully precompiled in 83 seconds. 193 already precompiled.

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
#Problem 1.1 - overall recycling and ash fractions

# % of total mass for each component
mass = [15, 40, 5, 3, 2, 5, 18, 4, 2, 2, 1, 3]

# Combustion ash (%) for each component
ash = [8, 7, 5, 10, 15, 2, 2, 100, 100, 100, 100, 70]

# MRF recycling rate (%) for each component
recycle = [0, 55, 15, 10, 0, 30, 40, 60, 75, 80, 50, 0]

# Overall recycling fraction (R) and ash fraction (A)
R = sum(mass .* recycle) / 10000      # 37.75% -> 0.3775
A = sum(mass .* ash) / 10000          # 16.41% -> 0.1641
```

```
println("Overall recycling fraction R = ", round(R; digits=4))
println("Overall ash fraction A      = ", round(A; digits=4))

# If you also want the recycled and ash masses for each city:
city_total = [100.0, 90.0, 120.0]    # Mg/day for cities 1, 2, 3

recycled_mass = city_total .* R
ash_mass      = city_total .* A

println("\nRecycled mass by city (Mg/day): ", round.(recycled_mass; digits=2))
println("Ash mass by city (Mg/day):      ", round.(ash_mass; digits=2))
```

```
Overall recycling fraction R = 0.3775
Overall ash fraction A      = 0.1641
```

```
Recycled mass by city (Mg/day): [37.75, 33.98, 45.3]
Ash mass by city (Mg/day):      [16.41, 14.77, 19.69]
```

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Here are the following decision variables for the municipal solid waste (MSW) management optimization model.

Variable	Meaning
$W_{ij} \geq 0$	Amount of waste (Mg/day) transported from city i to disposal facility j , where $j \in \{\text{LF}, \text{MRF}, \text{WTE}\}$. These variables represent the primary flows of waste from the cities to each facility.
$R_{kj} \geq 0$	Amount of residual waste or ash (Mg/day) transported from disposal facility k to facility j . This includes non-recycled residue from the MRF sent to LF or WTE, and ash from the WTE sent to LF. These variables represent secondary flows after processing.

Variable	Meaning
$Y_j \in \{0, 1\}$	Operational status of disposal facility j . $Y_j = 1$ if facility j is open (fixed costs and capacity apply), and $Y_j = 0$ if the facility is closed and cannot receive waste. Facilities include LF, MRF, and WTE.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

The objective is to minimize the total daily cost of the MSW system, including: 1. Transportation costs

2. Fixed operating costs for each facility

3. Variable tipping and processing costs

Variables

- $I = \{1, 2, 3\}$: set of cities
- $J = \{\text{LF}, \text{MRF}, \text{WTE}\}$: set of facilities
- $a = 1.5$: transportation cost (\$/Mg · km)
- l_{ij} : distance from city i to facility j
- l_{kj} : distance from facility k to facility j
- c_j : fixed daily operating cost of facility j
- b_j : variable cost per Mg at facility j

From the tables: - $c_{\text{LF}} = 2000$, $c_{\text{MRF}} = 1500$, $c_{\text{WTE}} = 2500$

- $b_{\text{LF}} = 50$, $b_{\text{WTE}} = 60$

- MRF effective variable cost:

$b_{\text{MRF}} = 7 + 40R$ with $R = 0.3775$

The total mass handled at facility j is

$$H_j = \sum_{i \in I} W_{ij} + \sum_{k \in J} R_{kj}.$$

Objective Function

The total transportation cost is

$$C_{\text{transportation}} = a \left(\sum_{i \in I} \sum_{j \in J} l_{ij} W_{ij} + \sum_{k \in J} \sum_{j \in J} l_{kj} R_{kj} \right).$$

The total fixed facility cost is

$$C_{\text{fixed}} = \sum_{j \in J} c_j Y_j.$$

The total variable processing cost is

$$C_{\text{var}} = \sum_{j \in J} b_j H_j.$$

Putting everything together, the complete objective is

$$\min_{W_{ij}, R_{kj}, Y_j} \left[a \left(\sum_{i \in I} \sum_{j \in J} l_{ij} W_{ij} + \sum_{k \in J} \sum_{j \in J} l_{kj} R_{kj} \right) + \sum_{j \in J} c_j Y_j + \sum_{j \in J} b_j H_j \right].$$

This matches the structure of the lecture formulation: transport costs, plus fixed facility costs, plus variable handling costs for each facility.

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Decision variables:

- W_{ij} : waste sent from city i to facility j (Mg/day)
- R_{kj} : residual or ash sent from facility k to facility j (Mg/day)
- Y_j : binary variable indicating if facility j is open (1) or closed (0)

with cities $i \in I = \{1, 2, 3\}$ and facilities $j, k \in J = \{\text{LF}, \text{MRF}, \text{WTE}\}$.

From Problem 1.1 we treat as **parameters**:

- overall recycling fraction $R = 0.3775$
- overall ash fraction $A = 0.1641$

Facility capacities (Mg/day):

- $K_{\text{LF}} = 200$
- $K_{\text{MRF}} = 350$
- $K_{\text{WTE}} = 210$

City waste generation (Mg/day):

- $S_1 = 100, S_2 = 90, S_3 = 120.$

1. City mass-balance constraints

All waste generated in each city must be sent to one or more facilities:

$$\sum_{j \in J} W_{ij} = S_i, \quad \forall i \in I.$$

These equations ensure that no waste is left unassigned or “created” at a city.

2. Facility capacity constraints

Total incoming material to each facility (from cities and other facilities) cannot exceed its capacity. We also link capacity to the binary variable Y_j so that if a facility is closed ($Y_j = 0$) it cannot receive any waste:

$$\sum_{i \in I} W_{ij} + \sum_{k \in J} R_{kj} \leq K_j Y_j, \quad \forall j \in J.$$

Here K_j is the capacity of facility j .

If $Y_j = 0$ the right-hand side is zero, forcing all flows to that facility to be zero. If there is any positive flow, it will be optimal to choose $Y_j = 1$ and incur the fixed cost in the objective.

3. MRF mass balance (recycling and residue)

Let the total mass entering the MRF be

$$H_{\text{MRF}} = \sum_{i \in I} W_{i,\text{MRF}}.$$

A fraction R of this flow is recycled and leaves the system, while the remaining fraction $(1 - R)$ becomes residue. All residue must go either to LF or to WTE:

$$R_{\text{MRF,LF}} + R_{\text{MRF,WTE}} = (1 - R) H_{\text{MRF}} = (1 - R) \sum_{i \in I} W_{i,\text{MRF}}.$$

This constraint enforces conservation of mass at the MRF: all non-recycled waste is accounted for in the residual flows.

4. WTE mass balance and ash generation

Let the total mass entering the WTE be

$$H_{\text{WTE}} = \sum_{i \in I} W_{i,\text{WTE}} + R_{\text{MRF,WTE}}.$$

A fraction A of this incoming mass becomes bottom ash that must be sent to the landfill. We represent this with the residual variable from WTE to LF:

$$R_{\text{WTE,LF}} = A H_{\text{WTE}} = A \left(\sum_{i \in I} W_{i,\text{WTE}} + R_{\text{MRF,WTE}} \right).$$

This constraint ensures that the ash produced by the WTE is consistent with the ash fraction computed in Problem 1.1 and that all ash is routed to the landfill.

5. Domain constraints

All flow variables must be nonnegative:

$$W_{ij} \geq 0, \quad \forall i \in I, j \in J,$$

$$R_{kj} \geq 0, \quad \forall k \in J, j \in J.$$

Facility status variables are binary:

$$Y_j \in \{0, 1\}, \quad \forall j \in J.$$

In practice, we only define R_{kj} for physically meaningful connections (e.g., MRF $\rightarrow$ WTE or LF, WTE $\rightarrow$ LF); all other R_{kj} are either omitted from the model or constrained to zero.

Together, these constraints enforce (i) conservation of mass at cities and facilities, (ii) consistency of recycling and ash generation with the fractions computed in Problem 1.1, and (iii) capacity and operational limits for each disposal option.

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in ‘data/generators.csv.’ In period 1, the demand is $d_1 = 1100$ MW. In period 2, the demand is projected to be $d_2 = 1200$ MW, but there is a 25% probability that it is \$1500. In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.